

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K})$ N : population at time t ; K : carrying capacity (depends on resources); $r, K > 0$.
Sol: $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \rightarrow K$ **Equilibrium points:** $N = 0$ (unstable), $N = K$ (stable). **Quickest:** $N = \frac{K}{2}$ ($<$: 凹; $>$: 凸)

Delay Model: Model: $\frac{dN}{dt} = f(N(t), N(t-T))$ T : delay time > 0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

¹Ext. of Logistic Model: $\frac{dN}{dt} = rN(t) [1 - \frac{N(t-T)}{K}]$ (用"ave. pop." 估计, 准用"Convolution Type") $\Rightarrow$ ²**Dimensionless:** $\frac{dN^*}{dt^*} = N^*(t^*) [1 - N^*(t^* - T^*)]$ ($N^* = \frac{N}{K}$, $t^* = rt$, $T^* = rT$)
^{1,2} **Periodic sol:** stable limit cycle $N(t + t_p) = N(t)$, t_p : period. **Amplitude:** rT **Equilibrium Points:** $N = 0 | N^* = 0$ (unstable); $N = K | N^* = 1$ (Stable: $0 < T^* = rT < \frac{\pi}{2}$; Unstable: $T^* > \frac{\pi}{2}$)

Continuous Population Model (EN): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$ $r, K, E > 0$; EN : harvesting rate/unit time; E : effort measure.

Equilibrium: $N = 0$ (unstable); $N = N_h(E) = K(1 - \frac{E}{r}) > 0$ (need $E < r$, Stable) **Yield:** $Y(E) = EN \cdot E > r$: die out; else: **Max Yield:** $Y_M = \frac{rK}{4}$ ($E = \frac{r}{2}$) **Steady State:** $N_h | Y_M = \frac{K}{2}$

Continuous Population Model (Y₀): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$ Y_0 : constant yield rate/unit time.

Max Yield: $Y_M = \frac{rK}{4}$ **Equilibrium:** $0 < Y_0 < Y_M$: Two positive points (unstable(小) and stable(大)). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of $N_{t+1} = f(N_t)$: Verhulst: $N_{t+1} = rN_t(1 - \frac{N_t}{K})$; $r, K > 0$. Better: **Ricker curve:** $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

Imperfect Bifurcation: $\dot{x} = rx - x^3 + h$ (codimension-2 bifurcation) If $r < 0$, x^* unique. If $r > 0$, $|h| > h_c$, x^* unique; $|h| < h_c$, x^* three. (两边 stable, un stable). Bifurcation points: $h = \pm h_c = \pm \frac{2r}{3} \sqrt{\frac{r}{3}}$. **cusp point:** $(r, h) = (0, 0)$

1.2 Summary - Stability and Bifurcation

	$\frac{dN}{dt} = f(N)$ No Limit cycle periodic solutions				S : stable, U : unstable.		$N_{t+1} = N_t F(N_t) = f(N_t)$ Eigenvalue: $\lambda = f'(N^*)$			
Stability	Stable: $f'(N^*) < 0$		Unstable: $f'(N^*) > 0$		Stable: $ \lambda < 1$		Unstable: $ \lambda > 1$ (chaotic)		Periodic: $ \lambda = 1 \begin{cases} + : \text{tangent bifurcation} \\ - : \text{period-doubling bifurcation} \end{cases}$	
Linearise	$N(t) = N^* + n(t), n(t) \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = \begin{cases} \text{(general)} = f(N^*) + f'(N^*)n(t) \\ \text{(delay)} = \text{直接带入 } t \text{ 后近似} \end{cases}$				Let $N_t = N^* + n_t, n_t \ll 1 \Rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \Rightarrow n_{t+1} = \lambda n_t$					
Bifurcation \\ Perturbations at N^* Periodic	Saddle-Node (Normal Form) $\dot{x} = r \pm x^2$ $0 \rightarrow S \rightarrow U-S$	Transcritical $\dot{x} = rx - x^2$ $S \rightarrow U \rightarrow S$	(super-)Pitchfork $\dot{x} = rx - x^3$ $S \rightarrow S-U-S$	Sub-critical Pitchfork $\dot{x} = r + x^3$ $U \rightarrow U-S-U$	$\lambda < -1$: $\uparrow$ oscillations		$-1 < \lambda < 0$: $\downarrow$ oscillations		$0 < \lambda < 1$: $\downarrow$ Monotone	
					If $fP(N^*) = N^*$, N^* is p -periodic.		$\forall$ even p -periodic N^* with $\lambda = -1$ branches into $2p$ -periodic		$\lambda > 1$: $\uparrow$ Monotone	
Other	Time-Independent: At N^* : $\frac{dN}{dt} = 0$ $N(t) = N(t-T) = N^*$ $f(N^*) = 0$				Sarkovskii's Theorem: If a 3-periodic N^* exists, then $\forall p \geq 1$, p -periodic N^* exists.					
Other	Tangency Condition: If $\dot{x} = f(x, r)$, 分岔点 (x^*, r_c) : $\frac{\partial f}{\partial x}(x^*, r_c) = 0$ $f(x^*, r_c) = 0$ 两点合并: coalesce .				Max/Min: (如果单峰 (unimodal)) 2-periodic $f'(N_m) = 0$, $N_{\max} = f(N_m)$; $N_{\min} = f(N_{\max})$					

Hopf Bifurcation: 平衡点从稳定变为不稳定, 同时产生 limit cycle. 条件: $\det(J) > 0$; $\frac{d\text{tr}(J)}{d\mu} \neq 0$ or: $\text{tr}(J)$ +/- 符号有变化 $\Leftrightarrow \lambda = \pm i\omega$ and $\frac{d\Re(\lambda)}{d\mu} \neq 0$ μ : parameter, not time.

$\Rightarrow$ Stable $\rightarrow$ Unstable + Stable Limit Cycle (supercritical) or Unstable $\rightarrow$ Stable + Unstable Limit Cycle (subcritical). (一般 super-) when $\text{tr}(J) = 0 | \Re(\lambda) = 0$, $\lambda = \pm i\omega \Rightarrow$ **periodic:** $T = \frac{2\pi}{\omega}$

Bifurcation: Qualitative changes in dynamics are called *bifurcations*. (当平衡点性质 | 个数发生变化时) **Bifurcation Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 | For Continuous Model: Find perturbation: $n(t)$ and $\frac{dn}{dt} \Rightarrow$ Assume $n(t) = e^{\lambda t} | ce^{\lambda t} \rightarrow$ Find $\lambda = g(\lambda) \Rightarrow$ Let $\lambda = u + i\omega \rightarrow u = g_1(u)$ and $\omega = g_2(\omega) \Rightarrow u < 0$ asy.stable; $u > 0$ unstable. **Find T_c :** (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check $T = 0$ stable. $\Rightarrow$ Using $u = 0$ (i.e. $\lambda = i\omega$) to find T_c .

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey): $\frac{dN}{dt} = N(a - bP)$; $\frac{dP}{dt} = P(cN - d) \Rightarrow$ nondim: $\frac{dN}{d\tau} = u(1 - v)$; $\frac{dP}{d\tau} = \alpha v(u - 1)$ **Equilibrium:** $(u^*, v^*) = (0, 0), (1, 1)$

Exact sol: $\alpha u + v - \ln(u^\alpha v) = H$, $H > H_{\min} = \alpha + 1$ constant. (closed phase) **Conservation Law:** Using $\frac{du}{dv} = \frac{du}{d\tau} \frac{d\tau}{dv}$ find $H \Rightarrow \frac{dH}{d\tau} = 0$, here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If $H(u, v)$ is level curve, Look $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix}_{(u^*, v^*)} \Rightarrow$ If +/- definite (i.e. $\forall \lambda_i > 0 (< 0)$)

$\Rightarrow$ It's local min/max at (u^*, v^*) & strictly convex/concave $\Rightarrow$ closed curves. Closed curves $\Rightarrow$ periodic sol.

Determinant Stability: For model: $\begin{bmatrix} u'(t) \\ v'(t) \end{bmatrix} = \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix}$ at equilibrium (u^*, v^*) . Let $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u - u^* \\ v - v^* \end{bmatrix}$, then $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix}_{(u^*, v^*)} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow$ Find eigenvalues λ of J by $\det(J - \lambda I) = 0$

¹ $\det(J) > 0$, $\text{tr}(J) < 0 \Rightarrow \Re(\lambda) < 0 \rightarrow$ stable node $\text{Im}(\lambda) = 0$; stable focus=spiral point $\text{Im}(\lambda) \neq 0$; ² $\det(J) > 0$, $\text{tr}(J) > 0 \Rightarrow \Re(\lambda) > 0 \rightarrow$ unstable node $\text{Im}(\lambda) = 0$; unstable focus=spiral point $\text{Im}(\lambda) \neq 0$

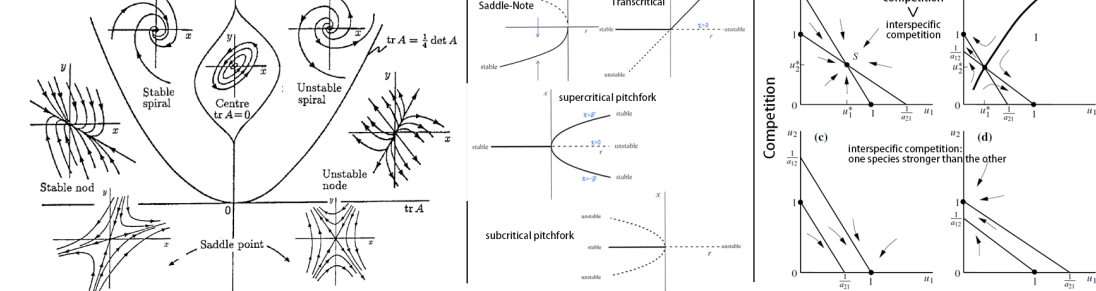
³ $\det(J) < 0 \Rightarrow \text{Im}(\lambda) = 0$, $\lambda_1 < 0 < \lambda_2 \rightarrow$ saddle point (unstable); ⁴ $\det(J) > 0$, $\text{tr}(J) = 0 \Rightarrow \lambda = \pm i\omega$ (pure imaginary) $\rightarrow$ center (only linear stability) | Hopf bifurcation (见前面)

Nullclines: $f(u, v) = 0$ or $g(u, v) = 0 \rightarrow$ 得 u, v 表达式是 nullclines. **Trapping Region:** A closed region R 可以用直线/nullclines 构成 s.t. 所有 边界的外法线方向 $\mathbf{n}$, 在对应边界上的点, 满足 $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

Find slope of Nullclines: $f(u, v) = 0 \Rightarrow df = f_u du + f_v dv = 0 \Rightarrow \frac{dv}{du} = -\frac{f_u}{f_v}$ Similarly: $\frac{dv}{du} = -\frac{g_u}{g_v}$

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in *trapping region* $\Rightarrow$ A **stable limit cycle** exists. (Cannot be a saddle (since $\det(J)$ cannot be < 0))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (b,c,d)



3 Reaction Kinetics

Enzyme Kinetics: e.g. $S + E \xrightleftharpoons[k_{-1}]{k_1} SE \xrightarrow{k_2} P + E$ K_i : rate of reaction; S : substrate; E : enzyme; SE : complex; P : product. **Rate of uptake:** $v = \frac{d[P]}{dt} |_{t=0}$

Law of Mass Action: Rate of reaction is proportional to the product of the concentrations of the reactants. e.g. $\alpha_1 A + \alpha_2 B \xrightleftharpoons[k_{-1}]{k_1} \beta_1 C + \beta_2 D \Rightarrow$ rateforward $= k_1 [A]^{\alpha_1} [B]^{\alpha_2}$; ratebackward $= k_{-1} [C]^{\beta_1} [D]^{\beta_2}$

Quasi-Steady State Approximation (QSSA): 如果系统里有快和慢变量, 则可以假设快变量达到稳态, 即 $\frac{d(\text{fast})}{dt} = 0$. (一般说来假设 complex concentration changes rapidly)

Singular Perturbations Analysis: For $\frac{du}{d\tau} = f(u, v)$, $\varepsilon \frac{dv}{d\tau} = g(u, v)$, $0 < \varepsilon \ll 1 \Rightarrow v$ fast (when $g \neq 0$), u slow. **Quasi-steady-state approximation:** $\varepsilon \rightarrow 0 \Rightarrow g(u, v) = 0$

Taylor: $u(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n u_n(\tau)$; $v(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n v_n(\tau)$ **Outer solution:** Taylor 带入原方程组 $\rightarrow$ 保留 $\mathcal{O}(1) | \varepsilon = 0$ Valid for $\tau > 0$ (slow time); $v_0(0)$ **invalid**. $\rightarrow$ not uniformly valid for $\tau \geq 0$

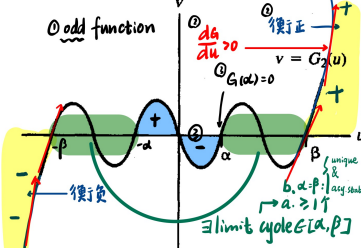
Inner solution: $\sigma = \frac{\tau}{\varepsilon}$, $V(\sigma; \varepsilon) = v(\tau; \varepsilon) \Rightarrow \frac{dV}{d\sigma} = \varepsilon f(u, v)$, $\frac{dV}{d\sigma} = g(u, v) \rightarrow$ Taylor 带入原方程组 $\rightarrow$ 保留 $\mathcal{O}(1) | \varepsilon = 0$ (known as **boundary layer** ($\tau = 0$ 处)); Valid $\tau = 0$ (fast time)

Asymptotic Matching: $\lim_{\sigma \rightarrow 0^+} [U_0(\sigma), V_0(\sigma)] = \lim_{\tau \rightarrow 0^+} [u_0(\tau), v_0(\tau)]$ **Uniformly Valid Solution:** $u(\tau; \varepsilon) = u_0(\tau) + \mathcal{O}(\varepsilon)$; $v(\tau; \varepsilon) = \begin{cases} v_0(\tau) + \mathcal{O}(\varepsilon), & 0 < \varepsilon \ll \tau \\ V_0(\sigma) + \mathcal{O}(\varepsilon), & 0 < \tau \ll 1 \end{cases}$

Activator-Inhibitor Model: For $\frac{du}{dt} = f(u, v)$, $\frac{dv}{dt} = g(u, v)$: If $\frac{\partial f}{\partial u} > 0$, u is an **activator** of v ; If $\frac{\partial f}{\partial v} < 0$, v is an **inhibitor** of u . ps: Possible have bifurcation behaviour

Positive/Negative Feedback: At Equilibrium: **Positive:** $f_u, f_v \geq 0$; $g_u, g_v \leq 0$ or $f_u, f_v \leq 0$; $g_u, g_v \geq 0$ **Negative:** $f_u, f_v \geq 0$; $g_u, g_v \geq 0$ or $f_u, f_v \leq 0$; $g_u, g_v \leq 0$

4 Relaxation Oscillations & Switch Behaviour



Find Limit Cycle: For $\begin{cases} \dot{u} = v - G(u) \\ \dot{v} = -u \end{cases} \Leftrightarrow \frac{d^2 u}{dt^2} + g(u) \frac{du}{dt} + u = 0$ with $G(u) = \int_0^u g(s) ds$. Lienard Equations If we have:

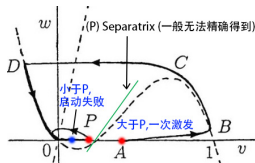
1: $G(-u) = -G(u)$ (odd) 2: $\exists \alpha > 0$ s.t. $G(\alpha) = 0$ and $G(u) < 0, \forall u \in (0, \alpha)$ 3: $\exists \beta \geq \alpha$ s.t. $G(u) > 0$, $\frac{dG}{du} > 0, \forall u > \beta$ and $G(u) \rightarrow \infty$ as $u \rightarrow \infty$.

Then: G is 'bumpy' function. 1 $\exists$ limit cycle between $(\alpha, \beta), (-\beta, -\alpha)$. 2 unique and asymptotically $\alpha = \beta$

Relaxation Oscillations: A periodic movement of "slow-fast alternation". e.g. For $\begin{cases} \varepsilon \dot{u} = v - G(u) \\ \dot{v} = -u | -u \varepsilon \end{cases}$ or More general form: $\begin{cases} \varepsilon \dot{u} = f(u, v) \\ \dot{v} = g(u, v) \end{cases} | \varepsilon \ll 1$

Relaxation Oscillations requies unstable equilibrium point. **Explain why it occur:** 1 Find *nullclines* (一般: 直线 + S 曲线) 2 Since $\dot{u} = \frac{\text{some func}}{\varepsilon} \gg 1$ (so that $|\dot{u}| \gg |\dot{v}|$) unless close to nullcline $\rightarrow$ (一般 S 曲线) 3 图上选 4 个点 ABCD, 一般选 S 的 local max/min 加 S 更远处两个点, 连接成一个 loop. 4 判断 ABCD 点的流向 (看 nullcline 两侧/上下) 5 分段分析: e.g. $A \rightarrow B \dot{u} = 0, \dot{v} = -u < 0$ 向下, 慢阶段. $B \rightarrow C$ not close to S curve, $|\dot{u}| \gg 1$ 向左, 快阶段/fast horizontal jump. $C \rightarrow D \dots D \rightarrow A \dots$

Estimate period T : 例如上图 (例子忽略了快反应) $T = \int_{t_A}^{t_B} dt + \int_{t_C}^{t_D} dt = 2 \int_{t_A}^{t_B} dt$ **Find $dt = _du$ 带入计算 e.g.** In slow time: $\dot{u} = 0 \rightarrow v = G(u)$ and $\dot{v} = -u \rightarrow \frac{dv}{dt} = \frac{dv}{du} \frac{du}{dt} = G'(u) \frac{du}{dt} = -u \rightarrow -\frac{G'(u)}{u} du = dt$, 带入计算



Excitable System: FitzHugh-Nagumo Model $\begin{cases} \frac{dx}{dt} = c[y + x - \frac{x^3}{3} + z(t)] \\ \frac{dy}{dt} = -x + a + by \end{cases}$ (x : excitability variable; y : recovery variable; $z(t)$: external stimulus)

Threshold 现象: **Small perturbation:** return to equilibrium (failed initiation/excitation). **Large perturbation:** large excursion before returning to equili. (action potential|successful initiation). After excitation| 大扰动之后: Refractory Period 不应期: system is unresponsive to further stimulation. **constant stimulus:** periodic firing (limit cycle).

Reason: 寻找 $\dot{u}, \dot{v}$ s.t. $\ll, \gg 1$ i.e. $|\dot{u}| \gg |\dot{v}|$ or $|\dot{u}| \ll |\dot{v}|$

Switch Behaviour: 从一个 stable point 到另一个 stable point. (前面 excitable system|relaxation oscillation 如有两 stable points 且画图分析 (像 relaxation oscillation 那样分析) 可得到)

5 Appendix

$y' + p(x)y = q(x)$: **Integrating Factor:** $I(x) = e^{\int p(x)dx} \Rightarrow I(x)y = \int I(x)q(x)dx$ **Taylor Expansion:** $f(x, r)$ at (x_0, r_0) : $f(x, r) = f(x_0, r_0) + (x - x_0)f_x|_{(x_0, r_0)} + (r - r_0)f_r|_{(x_0, r_0)} + \frac{1}{2} \dots$

Binomial Series: $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots$ **Linerisation:** $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$ $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \dots$ $\ln(a + x) = \ln(a) + \frac{x}{a} - \frac{x^2}{2a^2} + \frac{x^3}{3a^3} - \dots$

Hyperbolic Functions: $\sinh x = \frac{e^x - e^{-x}}{2}$ $\cosh x = \frac{e^x + e^{-x}}{2}$ $\tanh x = \frac{\sinh x}{\cosh x}$ $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$